Plan

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Two lectures on the OUU paper:

- Value-based uncertainty and multi-objective optimization
- Understanding orders and Functorial uncertainty

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Today:

- A few more slides on MOO and Functorial uncertainty

MOO: idealised problem (spending and emissions)

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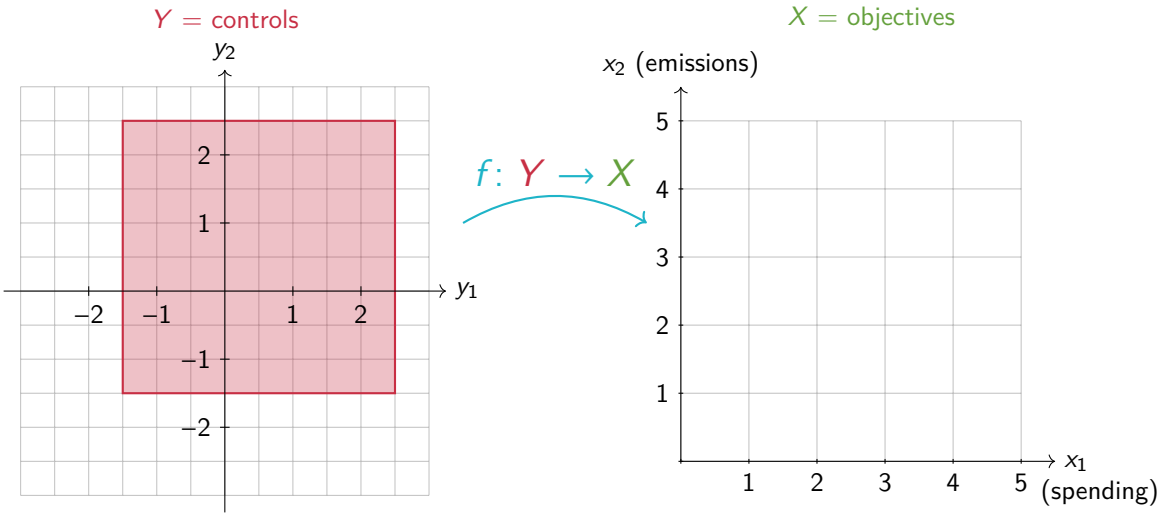

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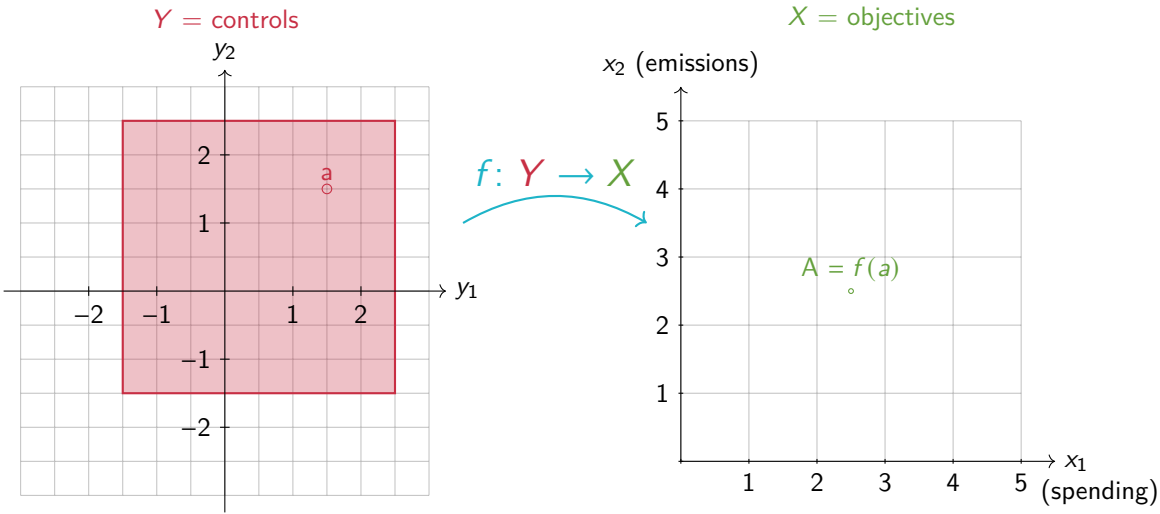

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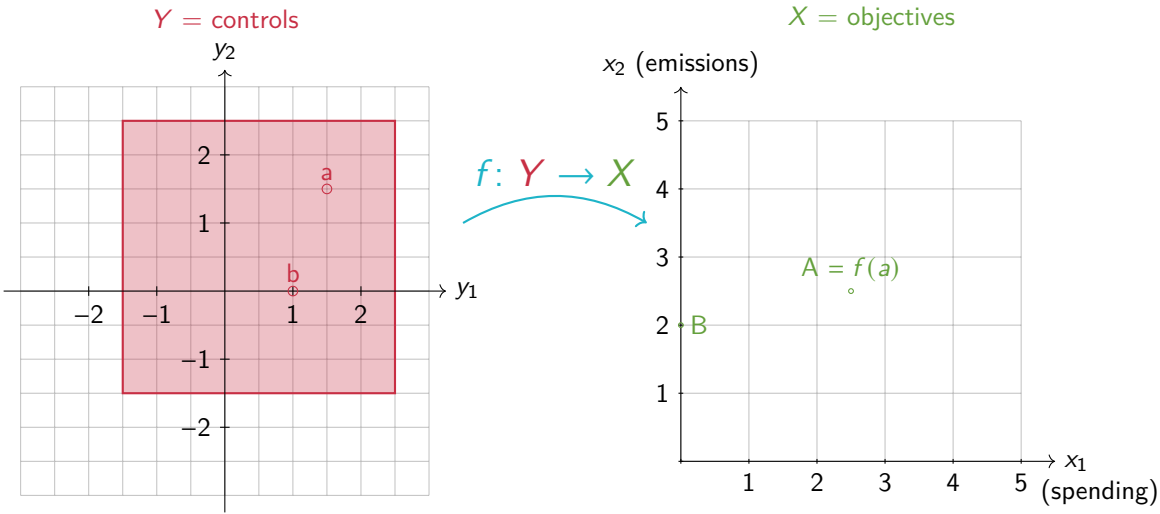

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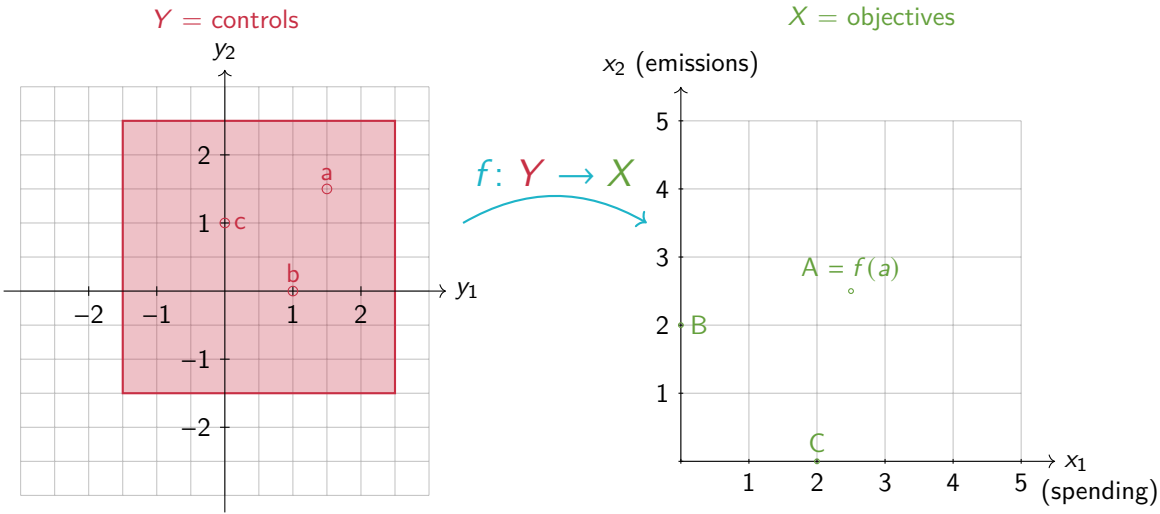

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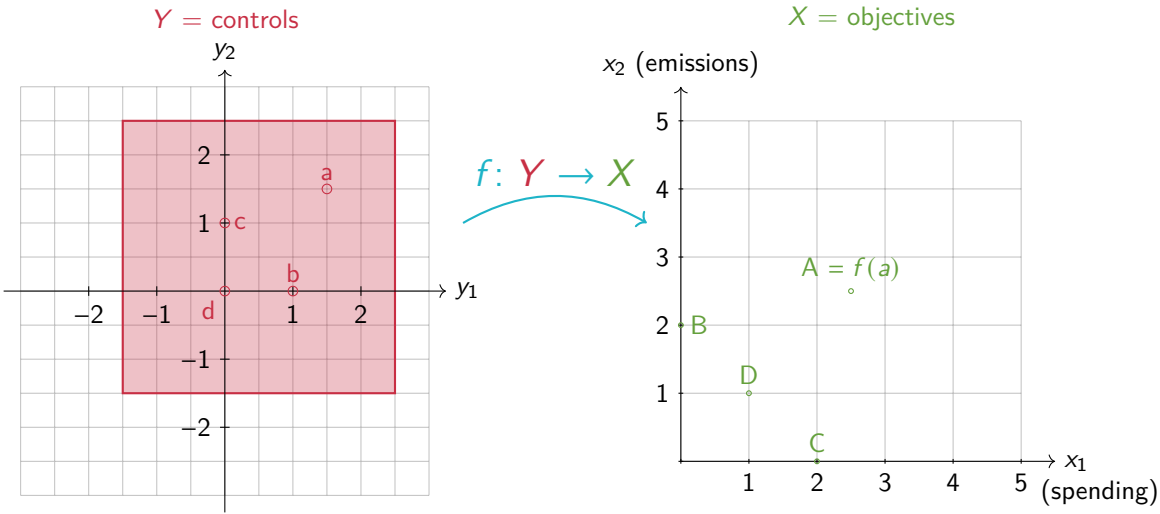

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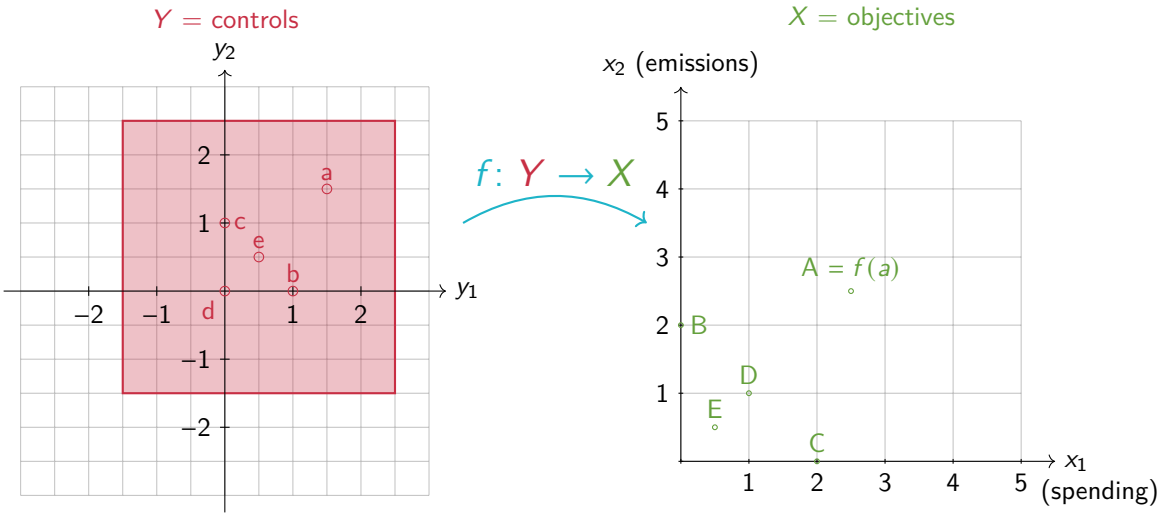

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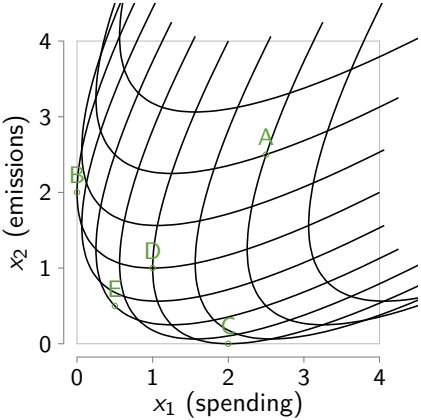
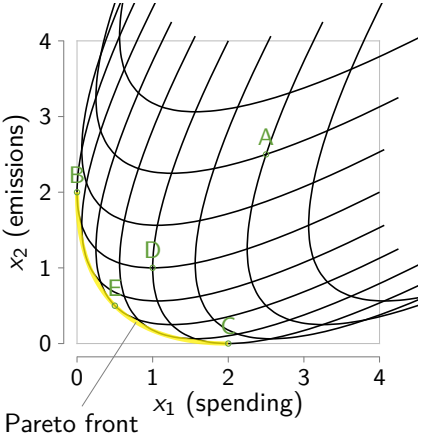


- Each curve keeps y_1 or y_2 fixed and varies the other control continuously.
- The curves shown are for $y_1 = -1, -0.75, \ldots, 1.75, 2$ (same for y_2).
- Here the objective function measures spending (x_1) and emissions (x_2).
- They can each reach zero, but not at the same time — there is a trade-off.

Point	y_1	y_2	spending	emissions	Note
A	1.5	1.5	2.5	2.5	uniformly bad
B	1	0	0	2	minimal spending
C	0	1	2	0	minimal emissions
D	0	0	1	1	bad compromise
E	0.5	0.5	0.5	0.5	**good compromise**

MOO: idealised problem (spending and emissions)



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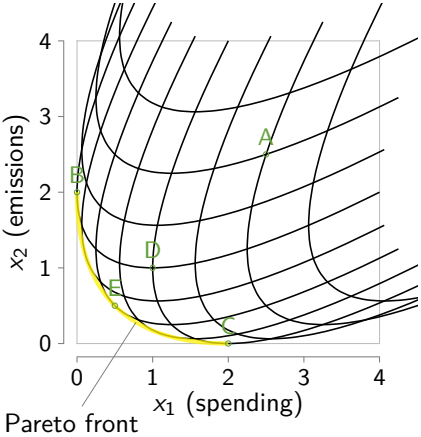

- Each curve keeps y_1 or y_2 fixed and varies the other control continuously.
- The curves shown are for $y_1 = -1, -0.75, \ldots, 1.75, 2$ (same for y_2)
- The Pareto front is the set of "optimal points" in the objective space.
- Aim: present the Pareto front to the policy maker ...
- ...or at least a few alternatives, like B, E, and C.
- When the objective space has more than 2 dimensions, approximating and visualizing the Pareto front becomes challenging.

Functorial uncertainty

Functorial uncertainty: finite probability distributions

- We have seen that values of type *List a* (*Set a*) can be used to represent non-deterministic uncertainty on outcomes of type *a*.

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$$\mathit{type}\ \mathit{SP}\ a\ =\ [\ (a, \mathbb{R}_+)\ ]$$

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$$\mathit{type}\ \mathit{SP}\ a\ =\ [\ (a, \mathbb{R}_+)\ ]$$

- A *SP a* represents a finite probability distribution on *a*:

$$\begin{array}{l} w :: \mathit{Eq}\ a\ \Rightarrow\ \mathit{SP}\ a\ \rightarrow\ a\ \rightarrow\ \mathbb{R}_+ \\ w\ d\ a\ =\ \mathit{sum}\ [p'\ \mid\ (a', p')\ \leftarrow\ d, a'\ ==\ a] \\ \mathit{prb} :: \mathit{Ord}\ a\ \Rightarrow\ \mathit{SP}\ a\ \rightarrow\ \mathit{Set}\ a\ \rightarrow\ \mathbb{R}_+ \\ \mathit{prb}\ d\ e\ =\ \mathit{sum}\ [w\ d\ a\ \mid\ a\ \leftarrow\ \mathit{map}\ \mathit{fst}\ d, \mathit{member}\ a\ e]\ /\ \mathit{sum}\ (\mathit{map}\ \mathit{snd}\ d) \end{array}$$

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- *prb d e* represents the probability of event *e* (a set of values of type *a*) according to *d*.

Functorial uncertainty: finite probability distributions

Exercise 3.1

Define *die* :: *SP Int* representing the probability distribution of a fair die. Apply

$$
\begin{aligned}
&\mathit{supp} :: \mathit{Eq}\ a \Rightarrow \mathit{SP}\ a \rightarrow [a] \\
&\mathit{supp}\ \mathit{sp} = \mathit{nub}\ (\mathit{map}\ \mathit{fst}\ \mathit{sp}) \\
&\mathit{ret} :: \mathit{Ord}\ a \Rightarrow a \rightarrow \mathit{Set}\ a \\
&\mathit{ret}\ a = \mathit{fromList}\ [a] \\
&\mathit{allEqual} :: (\mathit{Eq}\ a) \Rightarrow [a] \rightarrow \mathit{Bool} \\
&\mathit{allEqual}\ \mathit{xs} = \mathit{all}\ (\texttt{==}\ \mathit{head}\ \mathit{xs})\ (\mathit{tail}\ \mathit{xs})
\end{aligned}
$$

to implement a

$$\mathit{uniform} :: \mathit{Ord}\ a \Rightarrow \mathit{SP}\ a \rightarrow \mathit{Bool}$$

that checks for uniform, finite probability distributions. Apply *uniform* to check that your definition of *die* is fair. Apply *prb* to compute the probability that rolling the die obtains an outcome greater than 1.

Functorial uncertainty: finite probability distributions

- Modern probability theory was axiomatized by Kolmogorov in 1933, see `https://en.wikipedia.org/wiki/Probability_axioms`.

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Exercise 3.2

What is the sample space of values of type $SP\ x$? Apply Quickcheck and *prb* to check that your definition of *die* fulfils Axioms 1 and 2 of Kolmogorov's theory.

Functorial uncertainty: propagation of aleatoric uncertainty

- A deterministic system $X \to X$ can be iterated straightforwardly:

```
type DetSys a = a → a

detFlow :: DetSys a → Natural → DetSys a
detFlow f Zero     x = x
detFlow f (Succ n) x = detFlow f n (f x)

detTrj :: DetSys a → Natural → a → [a]
detTrj f Zero     x = x : []
detTrj f (Succ n) x = x : detTrj f n (f x)
```

Functorial uncertainty: propagation of aleatoric uncertainty

- A deterministic system $X \to X$ can be iterated straightforwardly:

$$
\begin{aligned}
&\mathit{type}\ \mathit{DetSys}\ a = a \to a \\
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&\mathit{detFlow}\ f\ \mathit{Zero}\ \ \ \ \ \ x = x \\
&\mathit{detFlow}\ f\ (\mathit{Succ}\ n)\ x = \mathit{detFlow}\ f\ n\ (f\ x) \\
&\\
&\mathit{detTrj} :: \mathit{DetSys}\ a \to \mathit{Natural} \to a \to [a] \\
&\mathit{detTrj}\ f\ \mathit{Zero}\ \ \ \ \ \ x = x : [\,] \\
&\mathit{detTrj}\ f\ (\mathit{Succ}\ n)\ x = x : \mathit{detTrj}\ f\ n\ (f\ x)
\end{aligned}
$$

Exercise 3.3

Define *detFlow* in point-free notation. Prove that $\mathit{detFlow}\ f\ n\ x = \mathit{last}\ (\mathit{detTrj}\ f\ n\ x)$ by induction on n and using the axiom $\mathit{last}\ (a : \mathit{as}) = \mathit{last}\ \mathit{as}$ (for nonempty as).

Functorial uncertainty: propagation of aleatoric uncertainty

- Non-deterministic systems $X \to [X]$ can also be iterated

```
type NonDetSys a = a → [a]

nonDetFlow :: NonDetSys a → Natural → NonDetSys a
nonDetFlow f Zero     x = [x]
nonDetFlow f (Succ n) x = concat (map (nonDetFlow f n) (f x))

nonDetTrj :: NonDetSys a → Natural → a → [[a]]
nonDetTrj f Zero     x = map (x :) [[]]
nonDetTrj f (Succ n) x = map (x :) (concat (map (nonDetTrj f n) (f x)))
```

Functorial uncertainty: propagation of aleatoric uncertainty

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&\\
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&\mathit{nonDetFlow}\ f\ \mathit{Zero}\ \ \ \ \ \ \ \ x\ =\ [x] \\
&\mathit{nonDetFlow}\ f\ (\mathit{Succ}\ n)\ x\ =\ \mathit{concat}\ (\mathit{map}\ (\mathit{nonDetFlow}\ f\ n)\ (f\ x)) \\
&\\
&\mathit{nonDetTrj} :: \mathit{NonDetSys}\ a \to \mathit{Natural} \to a \to [[a]] \\
&\mathit{nonDetTrj}\ f\ \mathit{Zero}\ \ \ \ \ \ \ \ x\ =\ \mathit{map}\ (x :)\ [[]] \\
&\mathit{nonDetTrj}\ f\ (\mathit{Succ}\ n)\ x\ =\ \mathit{map}\ (x :)\ (\mathit{concat}\ (\mathit{map}\ (\mathit{nonDetTrj}\ f\ n)\ (f\ x)))
\end{aligned}
$$

Exercise 3.4

Make sure that you understand the definitions of *nonDetFlow* and *nonDetTrj*. What are the types of $f\ x$, $\mathit{map}\ (\mathit{nonDetFlow}\ f\ n)$ and $\mathit{map}\ (\mathit{nonDetTrj}\ f\ n)$ in these definitions?

Functorial uncertainty: propagation of aleatoric uncertainty

Exercise 3.5

Let

$$\eta :: a \rightarrow [a]$$
$$\eta\ a = [a]$$
$$bind :: [a] \rightarrow (a \rightarrow [b]) \rightarrow [b]$$
$$bind\ as\ f = concat\ (map\ f\ as)$$

Redefine *nonDetFlow* and *nonDetTrj* in terms of η and *bind*.

Functorial uncertainty: propagation of aleatoric uncertainty

Exercise 3.5

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$$bind\ as\ f = concat\ (map\ f\ as)$$

Redefine *nonDetFlow* and *nonDetTrj* in terms of η and *bind*.

Exercise 3.6

Let *walk* :: *NonDetSys Int* be defined as

$$walk\ n = [n, n + 1\ ]$$

Compute the trajectories of *walk* of length *3* starting in *0* using both definitions of *nonDetTrj*.